

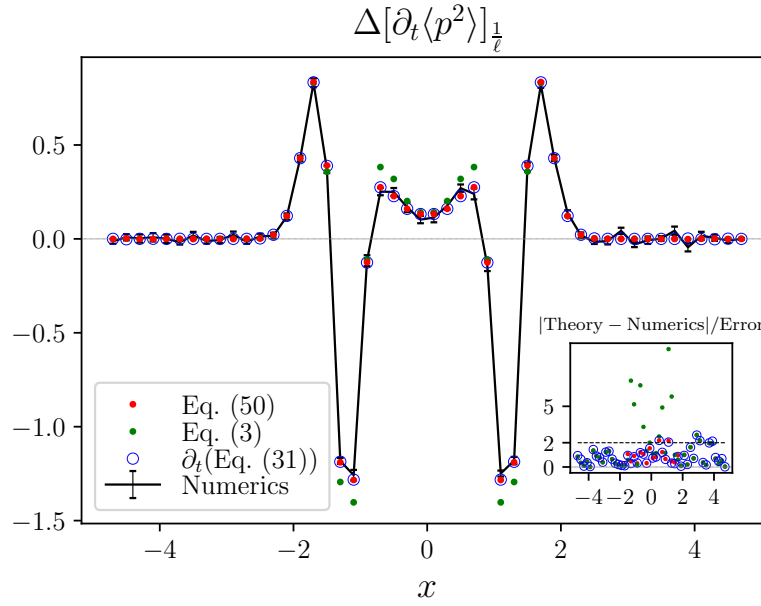


Figure 5: We show the $O(1/\ell)$ correction to time derivative of the second moment of hard rods velocity distribution from the initial state (59), as a function of space at $t = 1$. The black line represent the numerical simulation of the hard rods gas, for which the time derivative is evaluated through the difference quotient (61) with $\Delta t = 0.05$. Then, the $O(1/\ell)$ correction is estimated as the f_2 parameter of a fit with model $f(\ell) = f_1 + f_2/\ell$ to $\partial_t \langle p^2 \rangle(x, t; \ell)$, for $\ell \in \{500, 600, \dots, 1000\}$. The fit is performed independently for each point in space and the associated error bar is the standard deviation of the fit parameter. The numerical data are averaged over an ensemble of 8×10^{10} different realizations. The red dots represent the theoretical prediction of the new theory Eq. (50), while green dots are the prediction due to the Navier-Stokes-like theory Eq. (3). Finally, blue circles represent the time derivative of the solution (31), computed through difference quotient with $\Delta t = 0.01$. The inset shows the distance between theoretical predictions and data points in error bars units. One can clearly see that the new theory fits the numerical simulations much better, while the old Navier-Stokes-like theory Eq. (3) differs significantly. For numerical simulation we use the same data as in [1].

5 Conclusion

In this manuscript we derived the equation that governs the diffusive correction to the Euler (generalized) hydrodynamic description of the integrable hard rods model. The starting point for this is the exact microscopic solution formula (22). Assuming the usual thermodynamic/hydrodynamic large deviation scaling of correlation functions, we perform the large-scale $\ell \rightarrow \infty$ limit of (22), keeping the Euler $\sim 1/\ell^0$ and the diffusive contributions $\sim 1/\ell$. Then, taking a $t \rightarrow 0^+$ limit we finally obtain the equation governing the diffusive dynamics. We find that, unlike the previous Navier-Stokes like equation (3) [14], the equation for the one-point function up to, including, the diffusive scale, depends on the two-point function at the Euler scale. The equation for the Euler-scale two-point function depends on the Euler-scale one-point function, but not on higher order correlation functions (those can be neglected due to the assumed large deviation scaling). Therefore, the diffusive dynamics are described by a closed system of two coupled equations, instead of one, which can be solved